

Workshop 4 Instructions

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1 Introduction and set-up

In this workshop we're going to get hands-on practice working with data stored in three of the most common file types in our field: netcdf files, csv files, and ICARTT files, as well as how to work with/plot data from meteorological observations.

All of the data and the notebooks for this workshop are stored on a google drive as a zipped folder for you to download. Please go to the following link and download the zipped folder and then extract it to a location you prefer on your computer.

Click this link to be taken to the google drive folder containing everything you need.

Once you have downloaded and extracted the data the next thing to do is to create our python environment.

Enter the command below into your terminal to create a python environment called 'ws4_env' which contains everything you need for this workshop.

```
conda create --name ws4_env cartopy matplotlib metpy netcdf4 numpy pandas xarray
```

Once you've done that open up the 'Workshop-4.ipynb' notebook in your newly created environment to get started.

2 Instructions

1. Read and run the code in the 'Workshop-4.ipynb' notebook and get the basics of opening and working with data stored in files
2. Once you've gotten through that switch over to 'Workshop-4_Exercises.ipynb' and try to answer the questions in each section
3. Next start working through the 'Metpy_Basics.ipynb' notebook to learn how to use MetPy to work with sounding data along with other types of meteorological observations
4. If at any point you have questions feel free to ask JT or I for help, you should also feel free to work with the people around you

3 Additional Resources

This section contains additional resources if you're interested in learning more.

1. A tutorial/documentation guide for MetPy
2. An Introduction to a package called Siphon which lets you download meteorological data within your code
3. The Documentation for Xarray
4. The Documentation for Pandas